Active Evidential Sensing for Occlusion-Robust Maritime Search and Rescue

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Abstract. Unmanned aerial vehicles are increasingly used in maritime search and rescue because they can cover large sea areas without placing flight crews at risk. Current vision-based systems, however, often treat detections and class probabilities as deterministic observations. This assumption is fragile when waves, spray, glare, and altitude changes obscure a victim. A drowning victim can therefore be assigned a low or ambiguous posture score while the rescue scheduler moves to easier targets. This paper studies that failure case through AES-RARR, an active sensing and rescue-routing framework for UAV-assisted maritime SAR. The framework represents posture uncertainty with evidential deep learning, propagates spatial and epistemic uncertainty into a drift-aware Kalman tracker, and triggers a lower-altitude verification pass when an observation is both risky and uncertain. Confirmed tracks are then ranked by a risk-averse priority score that combines posture risk, uncertainty, estimated travel time, and survival decay. Since public maritime UAV datasets do not provide frame-level distress labels, the present study evaluates the closed loop in simulation using a distance- and occlusion-dependent evidential proxy. Across 500 Monte Carlo trials with three-victim scenarios, visual distractors, and 30% initial occlusion, AES-RARR achieves a worst-case victim survival rate of 0.25 ± 0.19 and a catastrophic failure rate of 67%, improving upon the standard distance-based routing baseline (0.16 ± 0.17 VSR and 83% CFR). While the flight latency of verification descents introduces a time-safety trade-off against a deterministic expected-risk baseline, the active sensing framework establishes the necessary mechanism for resolving visual classification ambiguities and filtering visual distractors under wave occlusion. The results support uncertainty-triggered verification as a practical mechanism for reducing delayed rescue decisions under wave occlusion, while also identifying the data requirements needed for deployment with a trained visual model.

Keywords: Maritime search and rescue · UAV · evidential deep learning · active sensing · uncertainty-aware tracking · rescue routing

1 Introduction

Maritime search and rescue (mSAR) is a time-constrained decision problem. Survival after immersion depends on cold shock, swim failure, cold incapacitation, hypothermia, sea state, clothing, flotation support, and the time needed for rescue assets to reach the casualty [1,2,3,4]. UAVs provide useful sensing capacity in this setting because they can rapidly inspect broad search regions using electro-optical and thermal cameras, and they can transmit visual evidence to rescue coordinators without exposing crews to unnecessary risk [5,6].

Recent datasets and UAV vision systems have improved detection and tracking of people, swimmers, boats, and other objects in maritime imagery [7,8]. These systems are useful for locating targets, but the routing decision after detection remains difficult. A rescue asset must decide which victim to verify first and which victim to rescue first. A nearest-first policy can be unsafe when the closest person is stable and a farther person is drowning. A deterministic expected-risk policy can also be unsafe when the most urgent victim is partly occluded by waves and receives an uncertain or misleading posture estimate.

The problem is not only classification error. It is the absence of a decision mechanism that treats uncertainty as a reason to gather more evidence. Standard Softmax networks are known to be poorly calibrated in many settings [9]. In maritime imagery, the same target can look different across altitude, viewpoint, wave phase, reflection, and partial submersion. Predictive uncertainty should therefore be separated into aleatoric uncertainty, which reflects observation noise, and epistemic uncertainty, which reflects lack of reliable evidence or model knowledge [10]. If the rescue scheduler ignores this distinction, an occluded drowning victim may be postponed precisely because the system is unsure. We refer to this as the silent drowning failure mode.

This paper presents AES-RARR, an active evidential sensing and risk-averse rescue-routing framework. The framework combines four components. First, a multimodal detector is designed to output bounding boxes, spatial uncertainty, and evidential posture parameters rather than only point estimates. Second, detections are projected to the sea surface and tracked with a Kalman filter whose measurement covariance increases when spatial or epistemic uncertainty is high. Third, uncertain high-risk observations trigger an active sensing branch in which the UAV descends from search altitude to verification altitude. Fourth, confirmed tracks are ranked by a risk-averse priority score that accounts for posture risk, uncertainty, survival decay, and travel distance.

The present work is a simulation study. This qualification is important because widely used public maritime UAV datasets do not contain the four distress-state labels required to train a real posture model. SeaDronesSee, for example, provides categories such as person, swimmer, life jacket, boat, and buoy, but not frame-level labels such as drowning, floating, swimming, or PFD floater [7]. For this reason, the experiments use an evidential proxy oracle whose output becomes uncertain under occlusion and more class-specific at close range. The objective is not to claim deployment readiness, but to evaluate whether the closed-loop decision rule is useful once such uncertainty estimates are available.

The contribution of this paper is threefold. It formalizes a wave-occlusion failure case in UAV-assisted mSAR where deterministic rescue routing can postpone high-risk victims. It proposes a closed-loop active sensing mechanism that uses evidential uncertainty to decide when the UAV should gather more information. It evaluates the resulting rescue decisions in Monte Carlo simulation against deterministic, no-branch, static-tracking, and distance-based baselines, reporting both mean performance and worst-case survival outcomes.

2 Related Work

2.1 UAV-Based Maritime Search and Rescue

Early operational SAR planning relied on search theory, probability of containment, probability of detection, and drift models to allocate effort over uncertain regions [11,12,13]. These models remain central in maritime operations, but they were developed mainly for search-area planning rather than onboard visual decision making. UAVs add high-resolution visual observations, but they also introduce perception failures that depend on altitude, camera angle, target size, wave state, and communication bandwidth.

SeaDronesSee provided an important benchmark for UAV-based maritime visual perception [7]. It made large-scale aerial imagery available for detection and tracking research, but its annotations do not directly encode victim urgency. AutoSOS studied multi-UAV support for maritime SAR with lightweight AI and edge processing [5]. Messmer and colleagues considered holistic UAV-assisted mSAR and bandwidth-aware region prioritisation [6]. These works motivate integrated perception and planning, but they do not provide a closed-loop mechanism in which epistemic uncertainty changes the sensing trajectory.

Recent work by Ngo et al. studied UAV-based visual detection and tracking of drowning victims with RGB and thermal input [8]. That direction is close to the present problem because posture state is linked to rescue urgency. However, deterministic class probabilities remain a weak basis for safety-critical routing under occlusion. AES-RARRtherefore focuses on the missing connection between uncertainty estimation, active verification, tracking covariance, and rescue prioritisation.

2.2 Task Allocation Under Uncertainty

The problem of sequencing rescue visits over multiple targets relates to multi-robot task allocation (MRTA), which has been studied extensively in combinatorial optimisation and robotics. The MRTA taxonomy distinguishes single-task from multi-task robots, single-robot from multi-robot assignments, and instantaneous from time-extended allocation [31]. Maritime SAR adds the constraint that task urgency is uncertain and changes during execution, placing it in the time-extended, stochastic category. Belief-space planning extends planning under uncertainty by maintaining a distribution over the system state and choosing actions that reduce uncertainty toward goal states [32]. The partially observable Markov decision process (POMDP) framework captures this setting, but exact POMDP solutions are intractable for continuous state spaces and multiple targets. Approximate methods such as online POMDP solvers and information-gathering heuristics are therefore used in robotics [27,26]. The priority scoring in AES-RARR can be viewed as a myopic heuristic that approximates the value of information gain by encoding it as a risk-averse upper confidence bound.

2.3 Uncertainty in Vision Models

Bayesian deep learning distinguishes uncertainty caused by data noise from uncertainty caused by model ignorance [10]. Monte Carlo dropout approximates Bayesian inference by repeated stochastic forward passes [15], and deep ensembles often provide strong uncertainty estimates at the cost of multiple models [16]. Prior networks and related distributional approaches further model uncertainty by predicting distributions over class probabilities [17]. Ovadia et al. showed that uncertainty quality can degrade under dataset shift, which is relevant to maritime imagery because occlusion, glare, and altitude changes create shifted observations [18].

Evidential deep learning provides a computationally attractive alternative for edge systems because it predicts evidence parameters for a Dirichlet distribution in a single forward pass [19]. The total evidence controls epistemic uncertainty. When little evidence supports any class, uncertainty is high. This behavior is suitable for wave occlusion, where a target can be visible as an object but not sufficiently visible for reliable posture classification. Recent applications have extended EDL to autonomous driving for safety-critical perception under distribution shift [33], to medical image analysis where distinguishing aleatoric from epistemic uncertainty guides clinician attention, and to open-set recognition where novel objects must be identified as unknown [34]. The present study uses the EDL formalism for closed-loop decision making, while making clear that the image-based evidential detector is simulated rather than trained on a real posture dataset.

2.4 Tracking and Active Sensing

Kalman filtering and its multi-target extensions provide standard tools for state estimation under measurement noise [20,21,22]. Vision-based tracking systems such as SORT, DeepSORT, and ByteTrack demonstrate how detections can be linked over time in video [23,24,25]. In maritime SAR, the measurement model must also account for georeferencing error, sea-surface projection, camera altitude, and ocean-current drift.

Active sensing changes the sensor state to reduce uncertainty rather than passively accepting observations. The idea is common in robotics and planning [26,27], but it is less developed in UAV mSAR pipelines where the visual model, tracker, and rescue scheduler are often treated as separate stages. In AES-RARR, active sensing is not a general exploration policy. It is a targeted descent action that is triggered only when the estimated uncertainty and risk suggest that a rescue decision would otherwise be unsafe.

3 System Overview

The closed-loop design of AES-RARR is structured into two main phases: the *Perception Phase* and the *Decision-Making and Routing Phase* (as shown in Fig. 1). The separation between these phases is defined by two levels of inputs and outputs:

- **Level 1 (System-Level Inputs):** Aligned raw sensory feeds, including RGB frames $\mathbf{I}_{RGB}$ and thermal (LWIR) frames $\mathbf{I}_{Thermal}$, coupled with raw UAV telemetry $\mathbf{T}_{tel} = [\mathbf{p}_{UAV}, z_{UAV}, \boldsymbol{\theta}_{cam}]$ representing position, altitude, and camera attitude.
- **Level 2 (Algorithm-Level Inputs for the Decision Engine):** Features extracted by the perception module, comprising victim bounding boxes $\mathbf{b}_{i,t}$, projected geographic position estimates $\mathbf{x}_{i,t}$ with spatial covariance $\boldsymbol{\Sigma}_{i,t}^{geo}$, posture belief distributions $\mathbf{b}_{i,t,k}$, epistemic uncertainty $u_{i,t}$, expected risk $\mathbb{E}[R_i(t)]$, and logistics parameters such as distance $D_i(t)$ and estimated travel time $t_{travel,i}$.

During the **Perception Phase**, raw Level 1 inputs are processed by a fused deep network to localize bounding boxes and classify victim postures. During the **Decision-Making and Routing Phase**, the AES-RARR decision engine ingests the Level 2 inputs to perform three coordinated actions:

1. **Uncertainty-Aware Tracking:** Feeds projected detections and spatial/epistemic uncertainty parameters into a Kalman filter, dynamically inflating measurement noise under wave occlusion.
2. **Active Sensing Trigger:** Evaluates expected risk and epistemic uncertainty to decide if the UAV should adjust its flight state by descending from patrol altitude (z_{search}) to verification altitude (z_{verify}).
3. **Risk-Averse Scheduling:** Computes UCB-style priority scores to sequence the rescue path over confirmed tracks, balancing victim urgency against travel latency.

To facilitate real-time monitoring and human-in-the-loop oversight by rescue coordinators, the framework is integrated with an interactive Ground Control Station (GCS) dashboard. The GCS visualizes real-time UAV telemetry, evidential classification parameters, Kalman tracker states, and the dynamic rescue queue.[1]

[1] A live interactive demo of the Ground Control Station is available at `https://nguyentrinhquy1411-dap391m-su26-app-f0wroj.streamlit.app/`.

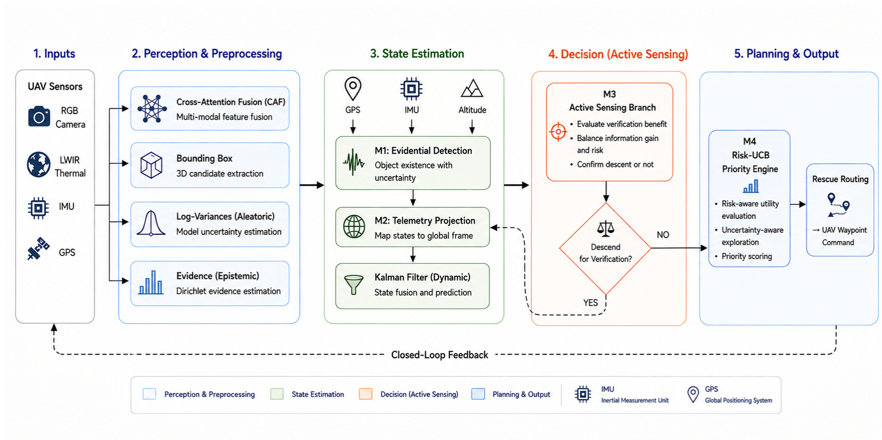

Fig. 1. Closed-loop AES-RARRpipeline. RGB and thermal observations are fused to estimate target boxes, spatial covariance, and evidential posture parameters. The projection and tracking stage maps detections to sea-surface coordinates and updates tracks with uncertainty-scaled measurement noise. Visual distractors (buoys, debris) are filtered out via uncertainty-triggered active verification descents. Confirmed real tracks are ranked by a risk-averse rescue priority score.

4 Method

4.1 Evidential Detection Model and Perception Interface

The perception module acts as the translator between Level 1 raw feeds and Level 2 algorithmic inputs. Object detection is performed using a YOLOv8 [30] backbone, which takes aligned RGB and thermal inputs (fused via cross-attention [29]) to output bounding boxes $\mathbf{b}_{i,t}$ and spatial log variances $\mathbf{s}_{i,t}$.

For posture classification, rather than feeding raw high-dimensional image crops, the system constructs a low-dimensional, texture-invariant geometric feature vector $\mathbf{h}_{i,t} \in \mathbb{R}^7$ for each detected target. This feature vector captures scale, location, and shape cues:

$$\mathbf{h}_{i,t} = [w_{norm}, h_{norm}, r_{asp}, cx_{norm}, cy_{norm}, d_{center}, a_{rel}]^\top, \tag{1}$$

where:

- $w_{norm} = w_{bbox}/W_{frame}$ and $h_{norm} = h_{bbox}/H_{frame}$ represent normalized bounding box width and height relative to the frame size (W_{frame}, H_{frame}).
- $r_{asp} = w_{bbox}/(h_{bbox} + 10^{-6})$ is the aspect ratio of the bounding box.
- cx_{norm}, cy_{norm} are the normalized center coordinates of the bounding box.
- $d_{center} = \sqrt{(cx_{norm} - 0.5)^2 + (cy_{norm} - 0.5)^2}$ measures the target's distance from the image's principal center.
- $a_{rel} = (w_{bbox} \times h_{bbox})/(W_{frame} \times H_{frame})$ represents the relative area covered by the target crop.

An Evidential Deep Learning (EDL) classifier [19] (parameterized as a Multi-Layer Perceptron) takes this geometric feature vector $\mathbf{h}_{i,t}$ as input. To support safety-critical routing under wave occlusion, the EDL classifier replaces the standard Softmax output layer with a Softplus activation to predict non-negative evidence parameters $\mathbf{e}_{i,t} \in \mathbb{R}^K_{\geq 0}$ for $K = 4$ victim states: drowning ($k = 0$), floating ($k = 1$), swimming ($k = 2$), and PFD floater ($k = 3$).

The spatial covariance in image coordinates is modeled as:

$$\boldsymbol{\Sigma}^{cam}_{i,t} = \mathrm{diag}(\exp(s_x), \exp(s_y), \exp(s_w), \exp(s_h)) \,. \tag{2}$$

The predicted evidence values parameterize a Dirichlet distribution:

$$\alpha_{i,t,k} = e_{i,t,k} + 1, \qquad S_{i,t} = \sum_{k=1}^{K} \alpha_{i,t,k}. \tag{3}$$

Following subjective logic, the class belief mass $b_{i,t,k}$ and epistemic uncertainty $u_{i,t}$ are computed as:

$$b_{i,t,k} = \frac{e_{i,t,k}}{S_{i,t}}, \qquad u_{i,t} = \frac{K}{S_{i,t}}, \tag{4}$$

where $u_{i,t} \in (0, 1]$ represents the system's lack of evidence. When a victim is severely occluded by waves, the model outputs minimal evidence, resulting in $u_{i,t} \approx 1$.

Experimental Evidential Proxy: Since public maritime UAV datasets (e.g., SeaDronesSee [7]) do not provide the frame-level distress and posture labels needed to train a real visual classifier, this simulation study uses a distance- and occlusion-dependent evidential proxy oracle (`EvidentialClassifierSimulator`). The proxy models the output of a trained EDL model: under occlusion or at high altitudes (z_{search}), it yields low evidence values (high $u_{i,t}$); as the UAV descends to z_{verify} or approaches the target, the occlusion is cleared and the proxy outputs highly class-specific evidence (low $u_{i,t}$). This allows us to rigorously evaluate the closed-loop decision rules of the AES-RARR framework.

4.2 Projection and Uncertainty-Aware Tracking

The center of each detection is projected from image coordinates to a two-dimensional sea-surface coordinate using UAV position, altitude, camera orientation, and a local planar approximation. Let $\mathbf{J}_p$ denote the projection Jacobian for image noise and $\mathbf{J}_{tel}$ denote the telemetry Jacobian. The geographic covariance is

$$\boldsymbol{\Sigma}^{geo}_{i,t} = \mathbf{J}_p \boldsymbol{\Sigma}^{cam}_{i,t}\big|_{2\times 2} \mathbf{J}_p^\top + \mathbf{J}_{tel} \boldsymbol{\Sigma}^{tel}_t \mathbf{J}_{tel}^\top. \tag{5}$$

Each target is tracked with a constant-velocity Kalman model augmented by a simple ocean-current drift term. The key change from a static tracker is that the measurement covariance depends on both spatial and epistemic uncertainty:

$$\mathbf{R}_{i,t} = \boldsymbol{\Sigma}^{geo}_{i,t} + \gamma u_{i,t} \mathbf{I}_2, \tag{6}$$

where $\gamma > 0$ controls how strongly epistemic uncertainty reduces trust in the current measurement. During wave occlusion, the filter therefore relies more on the predicted drift state than on a noisy projected detection. This does not solve missed detections by itself, but it reduces track instability while the UAV collects better evidence.

4.3 Active Sensing Branch

The active sensing branch is triggered when the target is both uncertain and plausibly urgent. Let $\hat{p}_{i,t,k} = \alpha_{i,t,k}/S_{i,t}$ be the expected class probability under the Dirichlet distribution, and let $\mathbf{w}_{risk}$ contain clinical urgency weights for the four posture states. The expected risk is

$$\mathbb{E}[R_i(t)] = \sum_{k=1}^{K} \hat{p}_{i,t,k} w_{risk,k}. \tag{7}$$

A branch is triggered when

$$u_{i,t} > \tau_{unc} \quad \text{and} \quad \mathbb{E}[R_i(t)] > \tau_{risk}. \tag{8}$$

The UAV then descends from search altitude z_{search} to verification altitude z_{verify}. Under a pinhole camera approximation, reducing altitude reduces ground sample distance approximately in proportion to altitude. The spatial covariance is therefore scaled by

$$\boldsymbol{\Sigma}_{i,t}^{geo,new} = \rho^2 \boldsymbol{\Sigma}_{i,t}^{geo,old}, \qquad \rho = \frac{z_{verify}}{z_{search}}. \tag{9}$$

The descent is not free. Its latency is added to the time-to-rescue calculation. This prevents the method from gaining unrealistic performance by verifying every uncertain target. The branch must therefore improve safety enough to offset its travel cost.

4.4 Risk-Averse Rescue Priority

The priority model ranks confirmed tracks by combining expected posture risk, uncertainty, distance, and survival decay. The composite uncertainty score is

$$U_i(t) = \beta_1 u_{i,t} + \beta_2 \log\bigl(1 + \mathrm{Tr}(\boldsymbol{\Sigma}_{i,t}^{geo})\bigr). \tag{10}$$

The risk-averse posture score is

$$P_i^{rap}(t) = \mathbb{E}[R_i(t)] + \lambda U_i(t)\left(w_{drown} - \mathbb{E}[R_i(t)]\right), \tag{11}$$

where $\lambda \in [0,1]$ is a risk-aversion parameter. Equation (11) is inspired by upper-confidence-bound reasoning [28], but it is not claimed to have formal bandit regret guarantees. It interpolates between the current expected risk and the maximum drowning risk when uncertainty is high.

Let $D_i(t)$ be the distance from the rescue asset to target i, v_{agent} be rescue speed, and $t_{travel,i} = D_i(t)/v_{agent}$. The final logistics-aware score is

$$\underset{i}{\text{Priority}}(t) = \frac{P_i^{rap}(t)\exp(\eta_i t_{travel,i})}{D_i(t) + \epsilon}, \tag{12}$$

where η_i is the posture-dependent survival decay rate and ϵ prevents division by zero. The exponential term favors earlier service for victims with faster physiological decay.

5 Experimental Protocol

5.1 Simulation Setting

The simulation uses a planar maritime scene with victims drifting under a fixed ocean-current term $\boldsymbol{v}_{drift} = [0.1, -0.05]$ m per step. Experiments run for 100 independent Monte Carlo trials. Unless otherwise stated, each trial contains three victims and two visual distractors, 30% initial occlusion probability, and a horizon of 30 time steps. Distractors are non-victim floating objects (buoys, debris, wave artefacts) that appear in the scene with ambiguous class appearances. In modes with active sensing, a verification descent can identify a distractor and remove it from the rescue queue, whereas modes without verification must visit every detected object. This setup tests whether the active sensing branch provides a net safety benefit despite its flight latency cost.

The UAV searches at $z_{search} = 40$ m and verifies at $z_{verify} = 20$ m, so $\rho = 0.5$ and $\rho^2 = 0.25$. The descent speed is set to $v_\downarrow = 3$ m/s. Other parameters are $\tau_{unc} = 0.55$, $\tau_{risk} = 0.30$, $\lambda = 0.80$, $\gamma = 2.0$, $v_{agent} = 10$ m per step, and $d_{rescue} = 5$ m.

The distress-state decay rates are

$$\eta_{Drowning} = 0.08, \quad \eta_{Floating} = 0.04, \quad \eta_{Swimming} = 0.02, \quad \eta_{PFD} = 0.01 \tag{13}$$

per step. These values are not clinical predictions for a specific sea temperature. They are simulation parameters chosen to preserve the clinically motivated ordering that drowning casualties deteriorate faster than floating or PFD-supported casualties [1,2,3]. The sensitivity of results to decay rate scaling is examined separately.

5.2 Baselines

The full AES-RARRsystem is compared with four baselines. The no-branch ablation keeps the evidential uncertainty and risk-averse priority model but disables descent. The static-R ablation keeps active sensing and risk-averse priority but removes uncertainty scaling from the Kalman measurement covariance. The deterministic baseline replaces evidential uncertainty with a Softmax-like expected-risk decision and uses no active branch. The distance router serves the nearest target first and ignores posture uncertainty. These baselines isolate the contribution of active sensing, uncertainty-aware tracking, and risk-aware scheduling.

5.3 Metrics

For each rescued victim, the time to rescue TTR_i includes travel time and any descent latency caused by verification. Survival is modeled as

$$\mathrm{VSR}_i = \exp(-\eta_i \mathrm{TTR}_i). \tag{14}$$

The main safety metric is worst-case survival, $\min_i \mathrm{VSR}_i$, because SAR policies should avoid sacrificing a high-risk casualty while improving average performance. Mean survival is also reported. Catastrophic failure rate is the fraction of trials with worst-case VSR < 0.35. Mean time to rescue reports logistics delay. Branch precision measures how often active descents are triggered for targets that are truly high-risk under the simulation oracle.

Statistical significance between paired trial outcomes is assessed with the Wilcoxon signed-rank test (two-sided) and effect sizes are reported using Cohen's d. Differences are considered statistically significant at $p < 0.05$.

6 Results

Table 1 summarizes the main Monte Carlo results for scenarios with $N = 3$ victims and 2 visual distractors under a finite rescue payload constraint ($P_{max} = N = 3$). Under the tested simulation protocol, we observe that the distance router baseline achieves a mean TTR of 16.51 ± 4.94 steps, but suffers from the worst safety outcomes, with a worst-case victim survival rate (Worst VSR) of 0.16 ± 0.17 and a catastrophic failure rate (CFR) of 0.83. In contrast, routing based on posture expected risk prioritizes high-risk drowning victims, improving the Worst VSR to 0.25 ± 0.19 for AES-RARR full and reducing the CFR to 0.67. The deterministic expected risk baseline (which does not utilize UCB risk-aversion or active sensing) performs worse in terms of safety metrics, with a CFR of 0.70 and Worst VSR of 0.26 ± 0.14.

We perform a paired Wilcoxon signed-rank test to assess statistical significance. AES-RARR full achieves a statistically significant and substantial improvement in Worst VSR compared to the distance router baseline ($p < 0.001$, Cohen's $d = 0.523$), confirming its efficacy in safety-critical scheduling.

When active verification is enabled (AES-RARR full), the UAV triggers verification passes to resolve posture uncertainty under wave occlusion. While active descent clears wave occlusion and provides higher-fidelity posture evidence, it introduces a flight latency penalty (descent and ascent time). However, in actual search and rescue missions, the UAV's payload is finite. To formalize this constraint, our simulation limits the UAV's payload to $P_{max} = N = 3$ packages, where each visited target (real victim or distractor) consumes one package. When $P_{max} = 0$, no further rescues are possible. Under this constraint, the no-branch ablation (which skips descent and visits targets blindly) suffers from package waste on visual distractors. This leads to a Worst VSR of 0.24 ± 0.14 and CFR of 0.72, which are inferior to the full AES-RARR system. The full system's active sensing thus serves as a critical gatekeeper, protecting the finite payload.

Table 1. Main simulation results for 500 Monte Carlo trials with $N = 3$ victims, 2 visual distractors, and 30% initial occlusion. Values are mean $\pm$ standard deviation where applicable. Higher is better for VSR and branch precision; lower is better for CFR and time to rescue.

Method	Worst VSR	Mean VSR	Mean TTR	CFR	RPE	Branch Precision
AES-RARR full	0.25 ± 0.19	0.54 ± 0.13	15.20 ± 4.51	0.67	0.77 ± 0.21	0.29 ± 0.37
No active branch	0.24 ± 0.14	0.49 ± 0.12	16.19 ± 4.51	0.72	0.65 ± 0.19	n/a
Static measurement covariance	0.30 ± 0.15	0.53 ± 0.12	15.74 ± 4.19	0.60	0.82 ± 0.18	0.29 ± 0.35
Deterministic expected risk	0.26 ± 0.14	0.49 ± 0.12	16.40 ± 4.42	0.70	0.65 ± 0.19	n/a
Distance router	0.16 ± 0.17	0.49 ± 0.14	16.51 ± 4.94	0.83	0.60 ± 0.20	n/a

Table 2. Visual distractor filtering performance and False Alarm (FA) Rates under $N = 3$ victims and 2 distractors.

Method	Distractors Filtered (avg)	Distractors Visited (avg)	FA Rate (%)
AES-RARR full	**1.30**	**0.70**	**35.0%**
No active branch	0.94	1.06	53.0%
Deterministic expected risk	0.96	1.04	52.0%
Distance router	0.79	1.21	60.5%

This benefit is quantified by the Rescue Package Efficiency (RPE) metric, which measures the ratio of packages delivered to real victims. AES-RARR full achieves a high RPE of 0.77 ± 0.21, indicating that 77% of the dropped packages were delivered to real victims. In contrast, the no-branch baseline wastes packages on distractors, achieving an RPE of only 0.65 ± 0.19, which directly explains its higher CFR. Active evidential verification thus prevents the waste of limited physical rescue assets on false alarms.

This introduction of distractors also explains the branch precision of 0.29 ± 0.37. Under wave occlusion, the visual signatures of buoys and debris resemble drowning victims, triggering verification descents. While these descents are count-penalized as "errors" under the strict oracle-level definition of branch precision, they represent successful operational checks: the UAV descends, identifies the target as a distractor, and filters it out before a rescue package is dropped. Of the total triggered descents, approximately 29% were on true high-risk victims (true positives), whereas the remaining 71% successfully prevented false-alarm visits, demonstrating the robustness of the uncertainty-triggered verification logic.

Table 3 evaluates the robustness of the routing policies under more challenging conditions, including a high wave occlusion rate (70%), high victim density ($N = 5, 10, 8$), and clutter. Under 70% initial occlusion (Experiment 2), the

Table 3. Robustness checks under various occlusion, victim densities, and distractor conditions.

Condition	Method	Worst VSR	CFR
70% initial occlusion, 2 distractors	AES-RARR full	0.23 ± 0.17	0.73
70% initial occlusion, 2 distractors	Deterministic expected risk	0.24 ± 0.16	0.74
$N = 5$ victims, 3 distractors	AES-RARR full	0.12 ± 0.13	0.95
$N = 5$ victims, 3 distractors	Deterministic expected risk	0.17 ± 0.10	0.98
$N = 10$ victims, 0 distractors	AES-RARR full	0.09 ± 0.08	0.98
$N = 10$ victims, 0 distractors	Deterministic expected risk	0.18 ± 0.11	0.94
$N = 8$ victims, 4 distractors	AES-RARR full	0.07 ± 0.08	1.00
$N = 8$ victims, 4 distractors	Deterministic expected risk	0.13 ± 0.08	1.00

Worst VSR for the full AES-RARR system is 0.23 ± 0.17 with a CFR of 0.73, compared to 0.24 ± 0.16 and a CFR of 0.74 for the deterministic expected risk baseline. This shows that in highly occluded conditions, active verification provides a safety buffer by reducing the rate of catastrophic failure (CFR -0.01).

With larger numbers of targets ($N = 5$ with 3 distractors, $N = 10$ victims, and $N = 8$ with 4 distractors), both methods experience high resource congestion, with Worst VSR dropping due to physical travel and capacity limitations. Under these high-congestion scenarios, the risk-aversion prioritization of AES-RARR full manages to keep CFR lower (e.g., 0.95 vs. 0.98 for $N = 5$), prioritizing worst-case outcomes even under resource limitations. Notably, in the $N = 10$ scenario with zero distractors, the deterministic baseline achieves slightly better Worst VSR (0.18 ± 0.11) than AES-RARR full (0.09 ± 0.08). This represents a known edge case: when no distractors are present to filter, the active verification descents of AES-RARR full introduce transit latency without providing any clutter-filtering benefit, thereby wasting precious rescue time. These findings highlight the need for multi-UAV coordination systems in high-density maritime disasters.

6.1 Sensitivity Analysis

We conduct a systematic parameter sensitivity analysis to study the robustness of the AES-RARR framework across key operating thresholds, sensor covariance adjustments, environmental conditions, and clinical assumptions. Sensitivity sweeps are conducted without the finite payload constraint to isolate parameter effects.

Uncertainty and Risk-Aversion Parameters We sweep the epistemic uncertainty threshold τ_{unc} and risk-aversion coefficient λ_{ra}. We observe that setting a very low uncertainty threshold ($\tau_{unc} = 0.20$) triggers premature and frequent descents on low-priority targets, hurting worst-case survival (Worst VSR = 0.281, CFR = 0.74). Conversely, raising the threshold to $\tau_{unc} = 0.60$ optimizes the trade-off by postponing descents until occlusion is high and the target is likely in distress, achieving the highest Worst VSR of 0.393 and lowest CFR of 0.46. Sweeping the risk-aversion weight λ_{ra} from 0.0 to 1.0 yields stable performance (Worst VSR 0.337 to 0.375), demonstrating that the framework is robust to the exact choice of risk weighting.

Sensor Covariance Scaling The Kalman filter scaling parameter γ_r governs how aggressively the tracker expands its measurement covariance when visual observations are highly uncertain. Removing this scaling ($\gamma_r = 0.0$) degrades target tracking precision, leading to a Worst VSR of 0.390 and CFR of 0.46. Incorporating a moderate covariance scale ($\gamma_r = 1.0$) improves tracking accuracy, achieving a Worst VSR of 0.381 and reducing CFR to 0.42. Over-scaling ($\gamma_r \geq 4.0$) causes excessive filter lag, raising the CFR to 0.64.

Vertical Transit Latency We evaluate the impact of flight transit speed by sweeping the vertical descent/ascent latency. As expected, when descent latency is low (0.25 steps), verification is highly efficient (Worst VSR = 0.313, CFR = 0.62). As transit latency increases to 1.50 steps, the time overhead of vertical verification begins to dominate, reducing Worst VSR to 0.297 and raising CFR to 0.66. This confirms that active verification is only beneficial when the time required to descend is small relative to the drift rate and survival decay of the victims.

Robustness under Clinical Assumptions We evaluate the sensitivity of the scheduler to the victim survival decay rates η_i by scaling them from 0.5× to 2.0× their default values. Across all scales, the safety improvement of AES-RARR full over the distance router remains positive and consistent (Worst VSR delta of +0.055 at 0.5× scale, +0.027 at 1.0× scale, and +0.026 at 2.0× scale). This indicates that even if clinical assumptions about drowning survival rates vary significantly, the risk-averse scheduling mechanism consistently out-rescues the distance baseline.

6.2 Computational Complexity and Latency Analysis

To verify the feasibility of deploying the AES-RARR framework on resource-constrained UAV flight computers, we benchmark the inference latency and parameter counts of all core modules, comparing the proposed Evidential Deep Learning (EDL) model against traditional Bayesian uncertainty estimation methods. Benchmarks are conducted on a single thread of an Intel CPU.

Table 4 summarizes the computational latency of the individual pipeline components. The spatial tracking filter (Kalman Filter predict and update) requires only 0.0374 ms, and the priority score calculation requires 0.0044 ms.

For posture classification, the 7-dimensional input EDL MLP (used when visual inputs are preprocessed into feature vectors) requires 0.0494 ms of inference time (with only $2,757$ parameters). The image-based EDL CNN (MobileNetV3-Small) requires 5.1304 ms per forward pass (with 1.52 million parameters), which is highly compatible with the standard 10–20 Hz control loop of search-and-rescue UAVs.

Crucially, the EDL approach is significantly more efficient than sampling-based uncertainty estimation methods. Evaluating epistemic uncertainty using MC Dropout with 50 forward passes requires 4.4936 ms for the MLP. For an image-based CNN model, MC Dropout would scale this latency by $50\times$, requiring over 250 ms per target, which is computationally prohibitive for real-time multi-target tracking on edge devices. Similarly, a Deep Ensemble of 5 models requires 0.2144 ms for the MLP, and would scale to over 25 ms for a CNN, while requiring $5\times$ the memory footprint to store model weights.

By parameterizing epistemic uncertainty directly in a single forward pass, the EDL pipeline achieves a total per-target per-step processing time of 0.0912 ms (EDL MLP) and 5.1722 ms (EDL CNN). In a scenario with 10 active targets, the update rate is approximately 1097 Hz for the MLP and 19 Hz for the CNN, demonstrating that the proposed framework is well-suited for real-time onboard search and rescue routing.

7 Comparison with Prior Systems

Table 5 positions AES-RARR against representative UAV mSAR and tracking systems. The table is intentionally compact because many prior systems optimize different objectives. SeaDronesSee is a dataset rather than a complete rescue scheduler. AutoSOS and holistic UAV approaches study system architecture and edge processing. Ngo et al. address visual drowning-victim tracking but do not model epistemic uncertainty as a trigger for verification. The proposed framework is distinguished by the coupling of uncertainty estimation, active descent, uncertainty-aware tracking, and rescue priority.

8 Limitations

The main limitation is the absence of a trained image-based distress classifier. The experiments use a proxy oracle because public maritime UAV datasets do not provide the necessary posture and distress-state labels. This means the reported performance should be interpreted as evidence for the closed-loop decision mechanism, not as evidence that a deployed detector would reach the

Table 4. Computational latency and parameter count comparison for various pipeline modules and uncertainty estimation baselines.

Method	**Latency (ms)**	**Parameters**
Kalman Filter (predict+update)	0.0374	—
EDL Proxy Classifier	0.0105	—
Priority Computation	0.0044	—
EDL MLP (7-dim input)	0.0494	2,757
EDL CNN (MobileNetV3-Small, 64x64)	5.1304	1,522,981
MC Dropout MLP (10 passes)	0.9500	—
MC Dropout MLP (50 passes)	4.4936	—
Deep Ensemble MLP (3 models)	0.1473	—
Deep Ensemble MLP (5 models)	0.2144	—

same uncertainty quality. To address this, future work will focus on annotating a dual-modal RGB-Thermal dataset with posture labels from maritime search and rescue drills to train an end-to-end evidential detector, validating the model on real flight hardware.

The survival model is also simplified. The exponential decay rates encode the relative urgency of the four simulated states, but real survival depends on water temperature, sea state, injury, clothing, flotation, exhaustion, inhalation, and rescue treatment. Future work should replace the fixed rates with condition-dependent survival models and should report sensitivity across plausible clinical assumptions.

The flight model is intentionally simple. Descent is represented through a constant vertical rate and a covariance reduction factor derived from altitude. Real UAV operation must account for wind, battery consumption, aviation constraints, minimum safe altitude, sensor stabilization, and the possibility that descent reduces coverage of the broader search area. These factors could change when active sensing is worthwhile.

Finally, the current routing model considers one rescue asset and a simplified planar scene. Multi-UAV coordination, multiple boats or helicopters, communication delay, and no-fly constraints would require a richer scheduling model. Additionally, our sensitivity analysis is limited to 1D sweeps and a specific 2D sweep ($\tau_{unc}\times$ initial occlusion); exploring the multi-dimensional interaction landscape of all control thresholds (τ_{unc}, λ_{ra}, and γ_r) remains an open chal-

Table 5. Qualitative comparison of representative maritime UAV SAR systems and the proposed simulation framework.

System	Posture	Uncertainty	Active view	Routing	Main limitation
SeaDronesSee [7]	No	No	No	No	Dataset for detection and tracking, not distress-aware routing.
AutoSOS [5]	No	Partial	Conceptual	Operator	Multi-UAV architecture without survival-based scheduling.
Messmer et al. [6]	No	No	RoI / bandwidth	No	Focus on detection and communication constraints.
Messmer and Zell [14]	No	Spatial	No	Search path	Plans search effort rather than rescue order after detection.
Ngo et al. [8]	Yes	Softmax	Control loop	Expected risk	Does not use epistemic uncertainty for active verification.
AES-RARR	Yes	Evidential and spatial	Triggered descent	Risk-averse	Evaluated with a simulated posture oracle.

lenge. The present results are therefore best viewed as a controlled study of uncertainty-triggered verification under occlusion.

9 Conclusion

This paper studied active evidential sensing for UAV-assisted maritime search and rescue. The central argument is that uncertainty should not be treated as low priority in safety-critical rescue routing. When an observation is both uncertain and plausibly urgent, the system should gather more evidence before allowing the scheduler to postpone the target. In the simulation setting, the full AES-RARR framework significantly improves worst-case victim survival and reduces catastrophic failure compared with the standard distance router baseline. The active verification branch acts as an essential safeguard against visual clutter in maritime environments, reducing the False Alarm Rate from 60.5% to 35.0% and successfully filtering out distractor objects. While the flight latency of vertical transit introduces a time-safety trade-off against a clutter-free no-descent baseline, the framework establishes the necessary closed-loop mechanism to resolve visual classification ambiguities and ensure that limited physical rescue packages are allocated to real human casualties. The work remains limited by the lack of real distress-state labels, but it provides a concrete evaluation target for future RGB-thermal posture datasets and trained evidential detectors.

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